GCWNet: A Global Context-Weaving Network for Object Detection in Remote Sensing Images

Yulin Wu, *Student Member, IEEE*, Ke Zhang, Jingyu Wang, *Member, IEEE*, Yezi Wang, Qi Wang, *Senior Member, IEEE*, and Xuelong Li, *Fellow, IEEE*

***Abstract*— With practical applications such as environment surveillance, agricultural production, and disaster assessment, accurate object detection in remote sensing images is in high demand. Precise detection of object instances in remote sensing images remains considerably challenging due to dense instance stacking, large-scale variations, and complex backgrounds. To solve the mentioned issues, a novel global context-weaving network (GCWNet) is developed for object detection in remote sensing images. We propose two novel modules for feature extraction and refinement, which include the global context aggregation module (GCAM) and the feature refinement module (FRM). GCAM assembles a global context with high-level and low-level features through feature weaving, which facilitates dense object detection. Meanwhile, FRM convolves multiple receptive fields by combining different branches, thereby further refining the features and improving the feature distinction at different scales. Furthermore, we design to alleviate the sample imbalanced problem during training using focal loss and balanced L1 loss to improve object classification and regression, respectively. The experimental results indicate that GCWNet achieves superior performance in object classification and localization on the DOTA-v1.5 dataset, which illustrates the superiority of GCWNet.**

***Index Terms*— Deep learning, feature enhancement, global context, object detection, remote sensing images.**

I. Introduction

With the launch of various satellites and the development of spaceborne sensors, remote sensing technology has made breathtaking progress. Object detection is a primary instrument for the interpretation and comprehension of remote sensing images [1], which is widely applicable to ecological environment monitoring [2], disaster forecasting and assessment [3], and marine environment investigation [4]. Over decades of research and development, although object detection has achieved better performance in natural images, there is still much room for improvement in object detection of remote sensing images because of considerable difference in characteristics with natural images.

Generally, the development of object detection has transitioned from the period of traditional object detection to deep learning period [5]. Before deep learning dominated the field of computer vision, the traditional detection algorithms [6] were the mainstream, which required manually designed features, with representative results such as histogram of oriented gradient (HOG) [7], Viola–Jones detectors [8], and deformable part-based model (DPM) [9]. Traditional object detection is relatively mature; however, it is difficult to design universal and robust features manually owing to the complexity and variability of image features. Deep learning has rapidly become the toast of object detection over recent years benefiting from powerful feature extraction and representation capabilities [10]–[12]. Specifically, a spurt of numerous classical deep-learning-based object detection frameworks have emerged, with these frameworks roughly categorized into one-stage and two-stage approaches. One-stage approaches [13] directly regress category and location of object from a segmented grid, which can be conducted in real time. The two-stage approach [14] requires generating numerous potential candidate regions, selecting the objects of interest based on these candidate regions, and then performing classification and regression, which generally have higher accuracy than the one-stage methods. The generic object detection methods have remarkable accomplishments on natural images through continued development. However, the results of these methods are not satisfactory when applied directly to remote sensing images.

While ground-based cameras capture natural images with a landscape perspective, remote sensing images are captured from a vertical perspective by a satellite or airborne camera at high altitude [15]. With the unique perspective of imaging, the object detection task in remote sensing images presents many challenges, primarily characterized by drastic scale variations, complex backgrounds, dense distributions, and lack of datasets. The details are as follows.

1) The variation in scale across categories is substantial as remote sensing images are derived from diverse spaceborne or airborne sensors and vary in time and space. Similarly, the reasons lead to a considerable difference in the scales of the objects in the same category.

Manuscript received January 4, 2022; revised February 10, 2022; accepted February 27, 2022. Date of publication March 2, 2022; date of current version April 5, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 61976179 and Grant U21B2041 and in part by the Innovation Capability Support Program of Shaanxi under Grant 2021KJXX-103. *(Corresponding author: Jingyu Wang.)*

Yulin Wu and Ke Zhang are with the School of Astronautics, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: wuyulin@mail.nwpu.edu.cn; zhangke@nwpu.edu.cn).

Jingyu Wang is with the School of Astronautics and the School of Artificial Intelligence, OPtics and ElectroNics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, China (e-mail: jywang@nwpu.edu.cn).

Yezi Wang is with the Structural Health Monitoring and Intelligent Structures Laboratory, Aircraft Strength Research Institute of China, Xi'an 710065, China (e-mail: kivib@mail.nwpu.edu.cn).

Qi Wang is with the School of Artificial Intelligence, OPtics and ElectroNics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, China (e-mail: crabwq@gmail.com).

Xuelong Li is with the School of Artificial Intelligence, OPtics and ElectroNics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, China, and also with the Key Laboratory of Intelligent Interaction and Applications, Ministry of Industry and Information Technology, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: li@nwpu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2022.3155899

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:58:37 UTC from IEEE Xplore. Restrictions apply.

2) The size of remote sensing images is large, and object instance usually occupies only a small portion of the image, with complex background information remaining. Furthermore, remote sensing images are susceptible to light, weather, and varying degrees of occlusion.
3) Since remote sensing images commonly cover a wide area, there are many small objects densely stacked in the scene, such as vehicles and ships (SHs).
4) Compared with natural image datasets, the publically released remote sensing image datasets developed for object detection tasks are limited due to prohibitive annotation costs.

Many researchers have proposed several approaches to tackle these crucial challenges. To suppress complex background and enhance feature representation, the solutions have been given from many perspectives. Li *et al.* [16] proposed an objectwise semantic representation with a newly introduced semantic segmentation module, which augments the holistic features. The saliency-guided single shot detector (SSD) (S-SSD) algorithm [17] integrated saliency information and guided the network to highlight information regions. Globally Aware road detection network with Multi-Scale residual learning Network (GAMSNet) [18] captured long-range dependences through multiscale residual learning, thus resolving road segments caused by complex backgrounds. With dense distribution of object instances, several approaches have attempted to address this problem. An anchor-free rotational SH detector named GRS-Det [19] and a rotational-region-based residual network (R3-Net) [8] are proposed for the detection of dense SHs and vehicles, respectively. Multi-category rotation Detector for Small, Cluttered and Rotated objects (SCRDet) [20] accomplished small and dense object detection using sample fusion network and attention network. Shi *et al.* [21] integrated global context features generated by the spatial attention mechanism on you only look once (YOLO) v3 to handle small object. Moreover, the multiscale detection issue is also a topical concern for researchers [22]. PG-YOLO [23] designed polymorphic modules to learn instance features of different scales and shapes simultaneously. Xu *et al.* [24] developed a hierarchical semantic dissemination mechanism to detect multiscale instances. Centerness-aware network (CANet) [25] embedded multiscale center descriptors into feature pyramid to separately generate central heat map to improve the feature discrimination of multiscale objects. Feature-merged single-shot detection (FMSSD) [26] proposed the feature merging module, which can aggregate multiscale context information. The existing approaches are generally task-specific and adopt different strategies to detect object instances in remote sensing images. Although these methods provide some solutions to the issues, there are still several problems ignored. First, global context could enhance object feature representation and assist in object classification and localization, which is currently not well-exploited. Second, most of the existing deep neural network receptive fields are fixed, which is not optimal for multiscale object instances. Finally, attributed to unbalanced sampling, the number of positive and negative samples varies greatly, and there is sample imbalance during the training process.

Based on the above observations, we propose a global context-weaving network (GCWNet) to address the object detection issues in remote sensing images. Abundant contexts are incorporated in the images, all of which are available to provide direction for generating more representative features. In this article, context mainly refers to high-level semantic context and low-level detail context. We design a global context aggregation module (GCAM) that aggregates high-level and low-level contexts, incorporating context into feature representation to achieve context-guided detection of dense objects. Moreover, normal convolution is limited by the fixed convolution kernel, and the receptive field is also fixed, which is less effective in processing multiscale objects. Therefore, we propose a feature refinement module (FRM) to enhance multiscale awareness, which focuses on refining further the features generated by GCWNet with convolution branches in different receptive fields. In addition, the instance region of interest (RoI) in the remote sensing image constitutes only a small fraction, mostly the background. Affected by complex backgrounds and sampling mechanism, the quantities of positive and negative patches are extremely unbalanced. Hence, the focal loss and the balanced L1 loss are introduced to mitigate the effect of sample imbalanced in both classification and regression. The main contributions of our work are as follows.

1) First, we propose a novel global context aggregation module named GCAM. The high-level and low-level context features are integrated with similar weaving, to improve feature discrimination and identify dense objects more accurately.
2) Second, for object instances involving drastic changes in scale and shape, we design feature refinement enhancement network, in which convolution layers of different receptive fields are combined to further refine and enhance multiscale object features.
3) Third, we newly introduce the focal loss and balanced L1 loss to alleviate the sample imbalance, which is caused by massive negative samples from complex backgrounds and sampling mechanisms.

The remaining sections of this article are organized as follows. In Section II, we briefly describe previous studies on generic object detection, object detection in remote sensing images, and context mechanisms. In Section III, we present the proposed GCWNet in detail. The experimental particulars and the results are available in Section IV. Finally, the conclusions are derived in Section V.

II. Related Works

A. Generic Object Detection

The ability to enable computational models to learn remarkably sophisticated, detailed, and advanced feature representations from data is the great benefit of deep learning. With the explosion and development of deep learning [27], generic object detection accuracy has arrived at a novel level. The existing object detection frameworks, depending on whether the candidate regions are generated or not, are classified as one-stage and two-stage frameworks. In the domain of the one-stage detection framework, all the locations in the image are considered as potential objects, and classification and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:58:37 UTC from IEEE Xplore. Restrictions apply.

regression are implemented directly at each location on the feature map. YOLO [28] and YOLOv4 [29] are representative one-stage methods with straightforward network structure and fast detection speed. The SSD [30] model performs detection on a single feature map of the feature pyramid, presenting a lower detection speed and slightly higher accuracy compared with YOLO. RetinaNet [31] combines feature pyramid and focal loss, which has improved for sample imbalance.

In contrast to direct classification and regression of the one-stage framework, the two-stage framework initially generates an extensive number of candidate regions and detects instances of interest from them. The regions with convolutional neural network (RCNN) [32] innovatively integrated AlexNet [33] with proposal selective search [34], but still used SVM classification. In the Fast RCNN, Girshick [35] proposed the RoI pooling layer to feed the proposal to the last convolution layer, thus enabling end-to-end training. As a progression of Fast RCNN, the Faster RCNN [36] implemented region proposal network (RPN) for proposal generation, which simplifies the calculation by setting anchor and consequently boosts the running speed. With the Faster RCNN as the basis, Lin *et al.* [37] suggested feature pyramid network (FPN) incorporating the lateral connections and the up–bottom pathway fusion features to facilitate multiscale object detection. Besides the classification and regression branches of Faster RCNN with FPN, Mask RCNN [38] supplemented with a parallel mask branch to accomplish instance segmentation. In addition, PANet [39] modified the feature pyramid with the addition of a bottom–up pathway to accommodate the instance segmentation task. In conclusion, with the additional proposal generation than the one-stage framework, the two-stage framework has higher accuracy but slightly inferior real-time performance.

B. Object Detection in Remote Sensing Images

Remote sensing image analysis and comprehension tasks [40], particularly object detection, are receiving ever more attention. The traditional methods of remote sensing object detection generally adopt manual features, producing very limited detection performance. For instance, Zhou *et al.* [41] designed an improved bag-of-words (BoW) model to detect vehicles in remote sensing images. The salient line section detector [42] was used to extract salient line sections from the input image and determine the airport area based on geometric features. In recent years, benefiting from the powerful feature representation of deep learning, a variety of effective deep learning methods [43] have been proposed and studied in remote sensing images. Using the one-stage framework YOLO, Xu and Wu [44] proposed a multireceptive fields' fusion network named MRFF-YOLO. Furthermore, refined SSD (RSSD) [45] obtained elementary predictions and then refined the network with RefinedNet to raise the confidence of the potential objects in remote sensing images. Based on RetinaNet, Yu *et al.* [9] proposed a cascaded rotating anchor-aided network to achieve high-accuracy detection of SHs.

The most popular detection framework is Faster RCNN with FPN [37], which also has numerous fascinating improvements for specific issues of remote sensing images. Partial improvement has been made to reorganize the contribution of features through attention mechanisms. Guo *et al.* [46] developed a hybrid method of scattering information enhancement (SIE) and attention pyramid network (APN) to perform aircraft detection in SAR images. Attention receptive pyramid network [47] and multiscale dense FPN [48] are also FPN-based variations. To balance multiple features on different layers at the semantic level, FBR-Net proposed by Fu *et al.* [49] applies an attention-guided balancing pyramid. SESPNet [50] used a route-level skip-connection architecture to augment the information flow, in addition to an attention module to calibrate the channel feature. Moreover, a part of the improvement is provided by feature fusion. The Feature Fusion Pyramid Network and deep Reinforcement Learning (FFPN-RL) [51] network combined feature fusion strategies and deep reinforcement learning. Zhang *et al.* [52] modified input data distribution by hard sample mining network, which enabled the network to learn the hard sample properties well. Multi-feature pyramid network (MFPNet) [53] incorporated two cascaded convolution pyramids to handle different scale instances in remote sensing images, while suppressing complex backgrounds with local context. Overall, researchers have made many modifications to the base model for the characteristics of bird's view images, but there is still room for improvement.

C. Contexts in Object Detection

Generally, the context features of an object refer to the features of its surrounding area, which may contain background or other adjacent object information [54], [55]. For instance, aircraft would only appear in the air or at airports, and storage tanks are usually surrounded by storage tanks as well. There are several attempts to implement contexts for object detection frameworks in both natural and remote sensing images. Specifically, for natural images, Chen *et al.* [56] suggested a hierarchical context embedding module to learn object-level conceptions using the overall image-level context. The multiscale contextual feature module [57] is inserted into the prediction layer to enrich the discriminative capability for multiscale detection. Leng *et al.* [58] improve the quality of candidate regions by context inference, and thus the robustness of small object detection. Furthermore, the recursive context routing (ReCoR) mechanism [59] is designed to encode context relationships and model contexts more efficiently.

Given the complex background and numerous small objects of remote sensing images, the context is more salient for remote sensing image object detection, and some representative applications have emerged. With respect to the scene context, Tao *et al.* [60] first segmented the road and the parking lot in the input image, and then detected vehicles in both the scenes separately. Similarly, scene-context FPNs (SCFPN) [61] are designed to reinforce the relationship between the object and the scene in remote sensing images. Based on local context, the context-aware convolutional neural networks (CA-CNNs) [62] explored adaptive extraction of local context features around potential regions and mapped them to multilevel feature maps. The context-driven detection network (CDD-Net) [63] used local contexts and hybrid attention pyramid to supplement the original features. Apart from that, Zhang *et al.* [64] used features of global and local contexts as well as attention modulation to construct the object detection network. Obviously, context features are remarkable

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:58:37 UTC from IEEE Xplore. Restrictions apply.

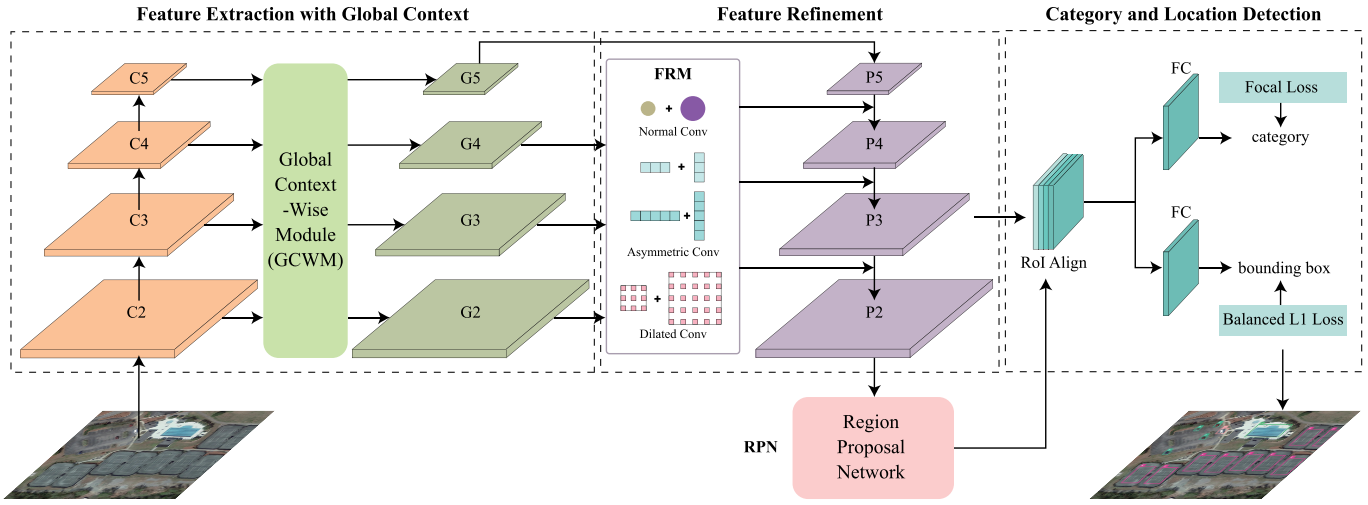


Fig. 1. Overall flowchart of the proposed GCWNet. For an input image, the features with global context are first extracted by the backbone network and GCAM. Next, the features are fed to the FRM for feature refinement to generate a feature pyramid. Subsequently, RPN is used to provide reliable proposals. Finally, the classification and regression branches with focal loss and balanced L1 loss classify and localize the object instances in the image, respectively.

for remote sensing object detection and worthy of more intensive investigation.

III. Methodology

The scale variation in remote sensing images is a prominent and distinct challenge for object detection, and therefore we aim to develop a novel framework which is effective for all scales. It is critical for the deep learning detection framework to extract features, and the discrimination of features directly impacts detection effectiveness. In this part, we implement enhanced feature representation with global context, including two proposed modules, namely, the GCAM and the FRM. In parallel, we resolve the category imbalanced during training using a novel loss function.

A. Architecture of GCWNet

Since the two-stage method has a competitive advantage in detection accuracy, we devise GCWNet based on the typical two-stage detection framework Faster RCNN with FPN [37]. According to the characteristics of remote sensing images, we have improved and innovated on the basis of the original framework. The pipeline of GCWNet consists of four stages: feature extraction with global context, feature refinement, region proposal generation, and category and location detection, as shown in Fig. 1.

1) Feature Extraction With Global Context: In the first stage, to obtain better extraction of remote sensing image features with complex backgrounds and multiscale objects, we adopt ResNet101 [65] as the backbone network. Then, to improve the differentiation of features and enhance features with global context, we design GCAM, which integrates semantic information from high level with low-level information. The GCAM achieves feature dimensionality consistency through upsampling of relatively higher level features and downsampling of comparatively lower level features, and fuses features through addition operation. The global context in GCAM is beneficial for the detection of dense objects.

2) Feature Refinement: Aiming to improve the perception of multiscale objects while reducing feature confusion from GCAM, we propose FRM. Motivated by Inception [66], our FRM uses a similar structure and combines asymmetric convolution and dilated convolution. The asymmetric convolution augments the perception of diverse shaped objects in remote sensing images, while dilated convolution increases the perception of multiscale objects by enlarging the receptive field. In addition, we preserve the original features through normal convolutional branches. Combining convolutional layers with different branches enables further refinement of the features and thus improves the detection efficiency.

3) Region Proposal Generation: Following the refinement of feature map, the next stage is to generate candidate regions, which is one of the characteristics of the two-stage network. We adopt RPN to implement the process of generating a large number of candidate regions. For capturing objects of various sizes and shapes, we predetermine a range of anchors having various scales and aspect ratios. The RPN achieves a sliding window by a 3×3 convolution, which is followed directly by two 1×1 convolutions for classification and localization, respectively. The output of the RPN is defined as the region of interest, which contains a series of candidate regions with high confidence.

4) Category and Location Detection: Due to the setting of anchors with different scales and aspect ratios, there are discrepancies in the size of the region of interest. To align different regions of interest and yield dimensionally consistent features, we adopt RoI Align. Then, the features are delivered to two fully connected layers for classification and localization. Simultaneously, nonmaximum suppression is used to prevent duplicate detection boxes. It is to be mentioned that the quantities of positive and negative samples are not balanced during the training process, and we use the focal loss and the balanced L1 loss to ameliorate the imbalance in regression and classification procedures, respectively.

B. Global Context Aggregation Module

Owing to the unique imaging properties of remote sensing images, there is abundant contextual information contained in the images. The valid context information is available to support the determination of whether a predefined category of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:58:37 UTC from IEEE Xplore. Restrictions apply.

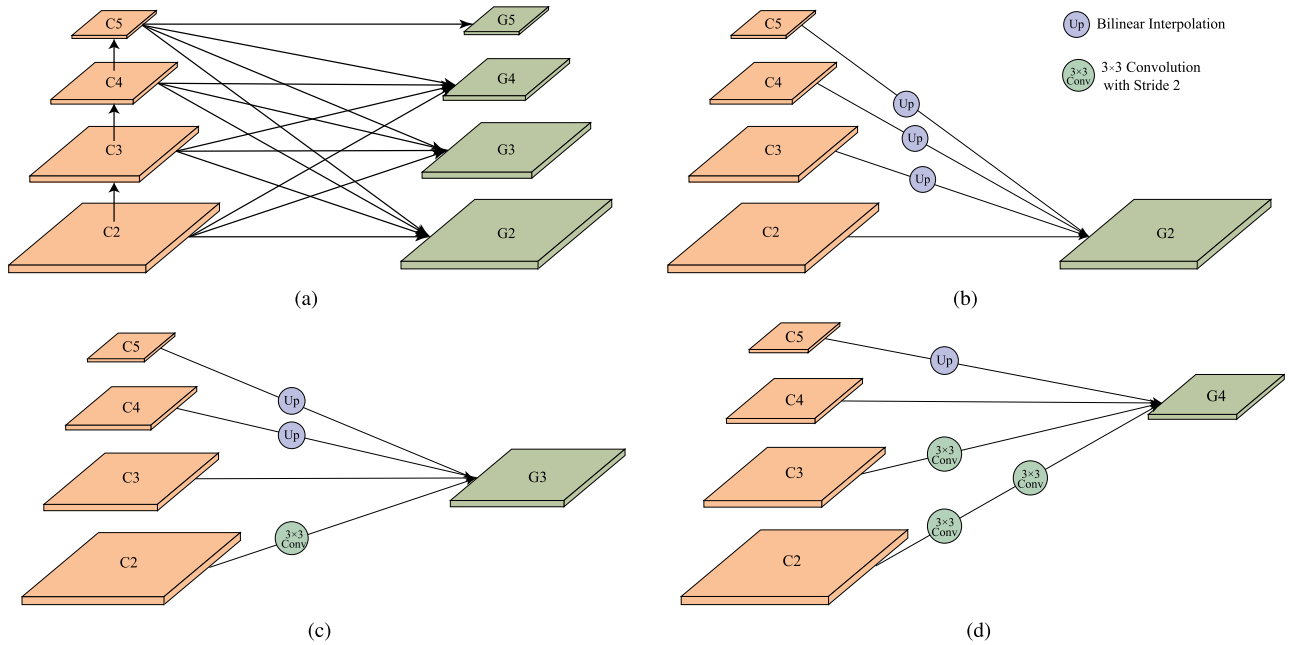


Fig. 2. Architecture and details of GCAM. It comprises three feature integration paths to implement high-resolution and low-resolution layer feature aggregation. (a) Overall illustration of GCAM. (b)–(d) Specific details of generating G2, G3, and G4, respectively. In addition, the path where the purple circle is located indicates upsampling with bilinear interpolation, and the path where the green circle is located indicates downsampling with one or two 3×3 convolutions.

objects exists at a specific location, essential for dense object detection. Moreover, Faster RCNN with FPN we selected is inadequate for context mining, and feature fusion is only accomplished through simple lateral 1×1 convolution and up–bottom upsampling. With these two motivations, we devise a GCAM to improve detection performance.

As shown in Fig. 2, GCAM sufficiently mines and fuses the context information through a novel feature fusion mechanism called feature weaving. Similar to FPN, we use ResNet101 as the backbone, while the output of the last layer of the residual blocks in Conv2, Conv3, Conv4, and Conv5 levels of ResNet101 is indicated as $\{C2, C3, C4, C5\}$. It is well-known that high-level feature maps such as $C5$ and $C4$ are smaller in scale, which contain more semantic features and are more beneficial for determining the object category. The low-level feature maps such as $C3$ and $C2$ are relatively large in scale and comprise more highly detailed features, which are more conducive to object localization. Thus, GCAM attempts to aggregate $C2$–$C5$ feature layers, and thereby integrates semantic and detailed features to obtain the global context. As shown in Fig. 2(a), the specific pattern for generating each Gi layer corresponding to Ci layer is as follows, with i as $\{2, 3, 4\}$. It is necessary to normalize the feature map size before feature fusion because the feature map size is not consistent in each layer. For lower level, higher level, and same-level features, we process the features by convolution layer downsampling, bilinear interpolation upsampling, and 1×1 convolution, respectively. In this step, channel dimension is modified to 256 uniformly. Finally, there is the consolidation of different layers with elementwise summation. Complementarily, based on HRNet [67], we perform downsampling with convolutional layers rather than simply scaling. For $2\times$ downsampling, a 3×3 convolution with stride 2 is used, while two sequential 3×3 convolutions with stride 2 are used for $4\times$ downsampling. Note that assuming $G5$ is derived in the same manner, $C2$ needs to be downsampled $6\times$, which introduces numerous parameters, and therefore a 1×1 convolution is used for $C5$ to generate $G5$ with the same dimensionality reduction to 256.

Considering the generation of $G3$ as an example, as shown in Fig. 2(c), the process of GCAM is as follows. The input features $\{C2, C3, C4, C5\}$ are initially fed to a 1×1 convolution layer for dimension reduction to 256. Next, $C5$ and $C4$ are bilinearly interpolated to upsample the feature map to reach the size of $C3$, while $C3$ is not subjected to upsampling or downsampling operations. Meanwhile, $C2$ is implemented as $2\times$ downsampling using a 3×3 convolution with stride 2. Finally, we add up the elementwise of all the four handled feature maps to obtain $G3$ with dimension 256. The same procedure is performed to align the feature map scales with $C2$ employing $C5$, $C4$, and $C3$ upsampling to generate $G2$, as shown in Fig. 2(b). $C5$ is upsampled, and $C3$ and $C2$ are downsampled by $2\times$ and $4\times$, respectively, which are combined with $C4$ to produce $G4$ as shown in Fig. 2(d). Similarly, the dimensions of $G2$ and $G4$ are 256. The operations of GCAM are calculated as follows:

$$\begin{cases} G2 = \text{Conv}(\text{Up}(C5)) + \text{Conv}(\text{Up}(C4)) + \text{Conv}(\text{Up}(C3)) \\ \qquad + \text{Conv}(C2) \\ G3 = \text{Conv}(\text{Up}(C5)) + \text{Conv}(\text{Up}(C4)) + \text{Conv}(C3) \\ \qquad + \text{Conv}(\text{Down}(C2)) \\ G4 = \text{Conv}(\text{Up}(C5)) + \text{Conv}(C4) + \text{Conv}(\text{Down}(C3)) \\ \qquad + \text{Conv}(\text{Down}(C2)) \\ G5 = \text{Conv}(C5) \end{cases} \tag{1}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:58:37 UTC from IEEE Xplore. Restrictions apply.

where Conv(·) denotes 1 × 1 convolution to align the channel numbers. Up(·) is the bilinear interpolation upsampling, and Down(·) indicates downsampling with 3 × 3 convolution.

C. Feature Refinement Module

To improve the perception of multiscale objects by solving the issue of fixed receptive fields of ordinary convolutional kernels, we develop the FRM to combine asymmetric convolution and dilated convolution by different branches. In [68], two parallel branches with kernels $1 \times d$ and $d \times 1$ are appended to normal convolution kernel $d \times d$, and then the outputs of the three paths are merged. For flipped images, it is extremely possible to achieve the same result as the input image with this setting at the axisymmetric position. In parallel, the rectangular receptive field of the asymmetric convolution kernel facilitates the recognition of objects with high aspect ratios. Instead of fusing asymmetric convolution kernels and normal convolution kernels to enhance skeleton features, we directly use asymmetric convolution of different shapes to reduce the calculation effort. The required downsampling operation in semantic segmentation degrades the image resolution and drops out information, whereby researchers have designed dilated convolution which extends the convolution filter with sparse parameters. Similarly, to enlarge the receptive field and strengthen the feature representation for multiscale instances, we also use dilated convolution.

The structure of FRM is illustrated in Fig. 3. The construction of FRM heavily depends on the concept of Inception. The FRM uses six parallel branches, with two normal convolutional branches, two asymmetric convolutional branches and two dilated convolutional branches, each of which consists of various kernel size convolution layers that span multiple receptive fields. To be specific, the first convolution layer of each branch is a 1 × 1 convolution devoted to converting the channel numbers. The first branch has only one convolutional layer, which could also be considered as a skip connection. The size of the convolution kernel in the second branch is 3 × 3, while asymmetric convolution kernels of 3 × 1 and 1 × 3 are used in the third branch. The fourth branch resembles the third branch with convolution kernel sizes of 5 × 1 and 1 × 5. Then, the fifth and sixth branches are constructed with 3 × 3 and 5 × 5 convolutional kernels with a dilation rate of 2, respectively. Eventually, feature maps with different receptive fields in all the six branches are combined in an elementwise summation to refine the features and enhance feature response of objects with diverse scales and geometries. For FRM, the output feature map of each branch and module is represented as follows:

$$\begin{cases} F1 = \{\mathrm{Conv}_{1\times1}(F_{\mathrm{in}})\} \\ F2 = \{\mathrm{Conv}_{1\times1}(F_{\mathrm{in}}), \mathrm{Conv}_{3\times3}(256)\} \\ F3 = \{\mathrm{Conv}_{1\times1}(F_{\mathrm{in}}), \mathrm{Conv}_{3\times1}(256), \mathrm{Conv}_{1\times3}(256)\} \\ F4 = \{\mathrm{Conv}_{1\times1}(F_{\mathrm{in}}), \mathrm{Conv}_{5\times1}(256), \mathrm{Conv}_{1\times5}(256)\} \\ F5 = \{\mathrm{Conv}_{1\times1}(F_{\mathrm{in}}), \mathrm{Conv}_{3\times3}(256)_{R=2}\} \\ F6 = \{\mathrm{Conv}_{1\times1}(F_{\mathrm{in}}), \mathrm{Conv}_{5\times5}(256)_{R=2}\} \end{cases} \tag{2}$$

$$F_{\mathrm{out}} = F1 \oplus F2 \oplus F3 \oplus F4 \oplus F5 \oplus F6 \tag{3}$$

where Fi denotes the different branches of the FRM. $\mathrm{Conv}_{k\times j}(n)_{R=2}$ stands for the convolutional layer with $k \times j$ convolutional kernels, n is the number of channels, and the dilation rate is 2. F_{in} and F_{out} indicate the input and output feature maps, respectively.

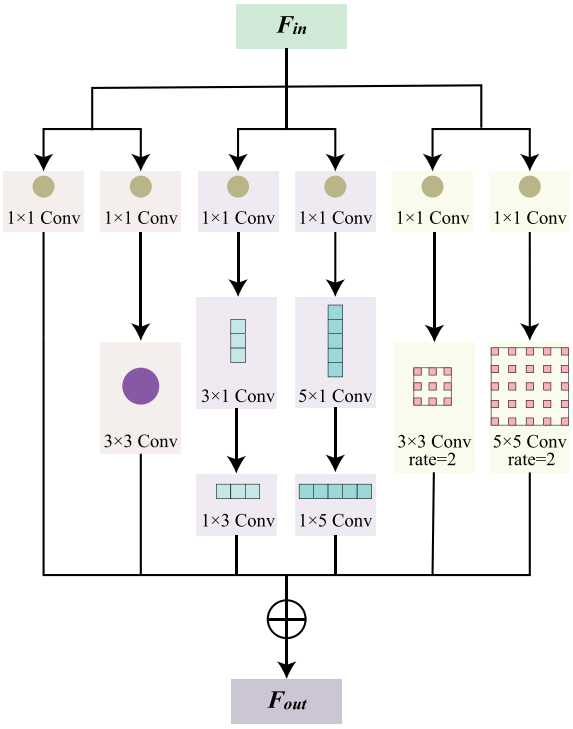


Fig. 3. FRM consists of two normal convolution branches, two asymmetric convolution branches and two dilated convolution branches combined. The normal convolution layer is used to maintain the original features, asymmetric convolution is applied to improve the awareness of different geometric objects, and dilated convolution is assigned to expand the receptive field, which boosts the detection of multiscale objects. The final output is calculated by elementwise summation of the cascade over all the branches.

In summary, we completely redesigned the lateral connection structure of the feature pyramid by incorporating GCAM and FRM. Specifically, the original features $\{C2, C3, C4, C5\}$ produce $\{G2, G3, G4, G5\}$ by distinctive feature weaving of GCAM, where $\{G2, G3, G4\}$ integrates the features of different layers and $G5$ contains only the features of the $C5$ layer. Since feature fusion is susceptible to feature conflation effect, it requires additional feature refinement, so $\{G2, G3, G4\}$ is fed into the FRM. Besides, $P5$ is obtained directly without any modification to $G5$. Similar to the original FPN, $P5$ is upsampled and summed with the output features after FRM refinement of $G4$, which yields $P4$. Then the operation is repeated to generate $P3$ and $P2$. The experimental results confirm that our modifications to the FPN help strengthen the feature representation and improve the detection efficiency.

D. Loss Function

The proposed GCWNet is designed on the basis of Faster RCNN with FPN, which is a two-stage object detection framework with end-to-end training. Similar to Faster RCNN with FPN, we adopt multitask loss to achieve minimization of the objective function, and the loss function is defined as follows:

$$L\left(p_i, l_i, t_i, t_i^*\right) = \frac{1}{N_{\mathrm{cls}}} \sum_i L_{\mathrm{cls}}(p_i, l_i) + \lambda \frac{1}{N_{\mathrm{reg}}} \sum_i p_i L_{\mathrm{reg}}\left(t_i, t_i^*\right) \tag{4}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:58:37 UTC from IEEE Xplore. Restrictions apply.

TABLE I
SIXTEEN CATEGORIES OF DOTA-V1.5 DATASET

BD	TC	SH	BC	PL	GTF	HA	BR
Baseball Diamond	Tennis Court	SHip	Basketball Court	PLane	Ground Track Field	HArbor	BRidge
SV	SBF	LV	HC	SP	ST	RA	CC
Small Vehicle	Soccer Ball Field	Large Vehicle	HeliCopter	Swimming Pool	Storage Tank	RoundAbout	Container Crane

where p_i is the probability distribution of the detected object category assessed by the softmax function, l_i indicates the label of the detected object, t_i represents the four parameterized coordinate vectors of prediction, and t_i^* shows the offsets of ground-truth and positive anchors. The hyperparameter λ determines the balance between the losses of the two tasks and follows the convention of setting to 1. Moreover, L_{cls} and L_{reg} denote the classification loss and regression loss and are normalized by N_{cls} and N_{reg}, respectively.

In the training procedure, the majority of anchors are marked as background or negative samples, and the minority of anchors are labeled as foreground or positive samples, thus resulting in the foreground–background imbalanced problem. We use focal loss [31] to overcome the imbalance between foreground–background samples. The focal loss concentrates attention on a few negative difficult samples rather than plenty of negative simple samples with (5) by adding a modulation factor γ to control the weight of each sample in the confidence loss

$$L_{\text{cls}}(p_i, l_i) = -(1 - l_i)^{\gamma} \log(1 - p_i) - (l_i)^{\gamma} \log(p_i). \quad (5)$$

In addition, we use the balanced L1 loss to address the issue of reducing imbalance in the regression process and increasing the localization accuracy as much as possible, which is formulated as follows:

$$L_{\text{reg}}(x) = \begin{cases} \dfrac{\alpha}{b}(b|x| + 1)\ln(b|x| + 1) - \alpha|x|, & |x| < 1 \\ \xi|x|, & \text{Otherwise} \end{cases} \quad (6)$$

where $|x|$ denotes $|t_i - t_i^*|$, and ξ refers to $\xi\alpha \ln(b + 1)$. In general, we separately set α to 0.5 and ξ to 2.

IV. EXPERIMENTS AND ANALYSIS

In this section, we first provide and describe the experimental dataset, evaluation metrics, and actually implementation details. Then, the ablation experiment is conducted to analyze the effect of the three portions of GCWNet on the performance. Finally, we show the comparison of GCWNet with the state-of-the-art algorithm to demonstrate the superiority of GCWNet.

A. Dataset and Evaluation Metrics

1) Dataset: DOTA [69] is publicly available for remote sensing image object detection known for its magnitude, which is considered as one of the most representative datasets. The dataset collected 2806 images from the various aerospace sensors and platforms. Each image is in the scale of 800 × 800–20 000 × 20 000 pixels and commonly comprises instances of various scales and shapes. The dataset is divided into DOTA-v1.0 and DOTA-v1.5 depending on the inclusion of instances of less than ten pixels. In this article, DOTA-v1.5 is used to develop and evaluate the proposed object detector GCWNet. There are 403 318 instances in DOTA-v1.5. Compared with DOTA-v1.0, there are 215 036 more instances, and the number of object categories has increased from 15 to 16. We provide all the object categories in Table I.

2) Evaluation Metric: Consistent with the majority of object detection approaches, we use the mean Average Precision (mAP) as an evaluation metric. mAP is a precise reflection of the global performance of object detection, which is defined as

$$\text{mAP} = \frac{1}{C}\sum_{i=1}^{C}\int_0^1 P_i(R_i), \quad dR_i \quad (7)$$

where C denotes the number of categories and it is set to 16 in this article. P_i and R_i represent the precision and recall of category i, respectively. $P_i(R_i)$ indicates the precision achieved when the recall of category i is R_i.

B. Implementation Details

1) Data Preprocessing: In general, remote sensing images are of high resolution and large scale. Owing to the limitation of computational resources, it is basically impossible to process the original image directly. Therefore, for the training and testing stages, we split the primary images into 1024 × 1024 patches with 100 pixels overlap between neighboring patches using the official toolkit. The overlap of 100 pixels is a tradeoff between the integrity of the instance and the number of images increased. All the experimental results are calculated based on the publicly available data only, since the test set of DOTA-v1.5 was released with images but not with ground truth. We crop and redivide the original training and validation sets into a novel 67% training set and 33% test set at random. To be emphasized, all the experimental results in this article are calculated on the repartitioned DOTA-v1.5 dataset.

2) Experiment Settings: We perform the designed algorithm with the broadly available pytorch1.5.1 framework and evaluate the algorithms on a TITAN RTX GPU with 24 GB of memory. We first initialize the network parameters with ResNet101 [65] pretrained on ImageNet for the purpose of accelerating convergence. During GCWNet learning procedure, the stochastic gradient descent (SGD) optimization algorithm is implemented to accelerate model training. The model is forcibly discontinued after 16 epochs of training. In parallel, the initial learning rate, weight decay, and momentum are fixed at 0.001, 0.005, and 0.9, respectively.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:58:37 UTC from IEEE Xplore. Restrictions apply.

TABLE II
ABLATION STUDIES OF MODEL COMPONENTS. THE HIGHEST VALUES IN EACH ROW ARE INDICATED BY BOLD NUMBERS

Settings	**mAP**	BD	TC	SH	BC	PL	GTF	HA	BR	SV	SBF	LV	HC	SP	ST	RA	CC
Baseline [37]	57.3	70.1	86.0	40.3	69.4	78.6	68.5	59.5	55.1	23.7	61.1	45.4	68.3	64.5	46.4	56.2	24.4
+ GCAM	59.6	67.8	82.8	33.8	**72.9**	79.4	**71.9**	68.2	**63.5**	24.3	64.0	48.7	67.5	66.6	47.9	**59.8**	33.6
+ FRM	59.4	72.6	87.2	53.1	68.7	80.9	67.7	67.5	53.2	25.3	61.0	50.1	68.4	66.8	50.6	53.2	23.9
+ Loss	58.5	67.9	81.6	40.2	71.6	79.7	69.4	68.4	63.1	24.3	64.8	47.4	61.4	65.1	42.0	58.2	31.1
GCWNet	**62.7**	**77.3**	**89.7**	**53.7**	71.4	**81.3**	70.8	**70.3**	55.7	**26.0**	**65.7**	**54.2**	**72.2**	**69.0**	**52.6**	59.6	**33.6**

TABLE III
ABLATION STUDY OF THE COMPONENT FEATURE LAYERS OF GCAM

Settings	**mAP**	BD	TC	SH	BC	PL	GTF	HA	BR	SV	SBF	LV	HC	SP	ST	RA	CC
Baseline [37]	57.3	70.1	**86.0**	**40.3**	69.4	78.6	68.5	59.5	55.1	23.7	61.1	45.4	**68.3**	64.5	46.4	56.2	24.4
{C3,C4,C5}	58.4	66.4	80.2	33.2	70.5	78.7	71.7	67.8	63.0	24.2	62.9	46.8	67.3	65.4	43.1	59.2	34.0
{C2,C4,C5}	58.7	68.2	81.1	33.5	71.7	78.3	71.6	67.9	62.9	24.2	63.1	46.7	67.9	65.3	41.9	61.5	33.2
{C2,C3,C5}	59.0	66.0	82.6	33.7	72.2	79.0	71.1	67.5	62.7	24.3	63.8	48.5	67.8	66.0	43.8	60.6	**34.1**
{C2,C3,C4}	59.1	**71.0**	81.5	33.5	71.7	78.4	71.7	67.9	63.4	24.2	63.9	47.7	67.2	65.3	42.7	**61.6**	33.5
+ GCAM	**59.6**	67.8	82.8	33.8	**72.9**	**79.4**	**71.9**	**68.2**	**63.5**	**24.3**	**64.0**	**48.7**	67.5	**66.6**	**47.9**	59.8	33.6

C. Ablation Evaluation

To verify the contribution of our proposed three modules, GCAM, FRM, and Loss, to the performance improvement of the GCWNet model, we conducted ablation experiments to research different network models, which primarily include the following.

1) Baseline: We adopt Faster RCNN with FPN as the baseline network, and our network setup remains the same as [37] unless otherwise stated.

2) Baseline + GCAM: The baseline with GCAM only is inclusive. The FRM is replaced by a 1×1 convolution in Fig. 1. The loss is consistent with the baseline.

3) Baseline + FRM: We substitute the original horizontal connection with FRM on the basis of baseline.

4) Baseline + Loss: The focal loss and the balanced L1 loss are set as the corresponding classification loss and regression loss in the baseline, respectively.

5) GCWNet: We implement the proposed global contextual aggregation network comprehensively, assembling GCAM, FRM, and loss modules on the baseline.

Table II provides the results of the different models in the ablation experiments in detail for each category. It is evident from Table II that the mAP of the baseline model reaches only 57.3%, while all our proposed modules are able to improve the detection accuracy to varying degrees. With the addition of GCAM, mAP gains 2.3% boost and achieves the best performance on four categories, indicating that the global context aggregation significantly optimizes the features. The baseline + FRM yields a 2.1% improvement in mAP, which demonstrates the effectiveness of expanding the receptive field in remote sensing image object detection. Meanwhile, the mAP increase by 1.2% is achieved by modifying the loss function of the model, which shows the necessity of ameliorating the imbalanced in the training process. Eventually, GCWNet shows the superiority of mAP compared with the baseline by increasing from 57.3% to 62.7%, which is higher than adding either module individually.

From the perspective of each category, both the proposed individual modules and GCWNet exhibit enhancement compared with the baseline. For instance, there is a 13.4% and 10.8% rise in mAP for SH and harbor (HA), respectively, which indicates that GCWNet successfully enhances the detection effectiveness of instances with large aspect ratio. Furthermore, small vehicle (SV) and soccer ball field (SBF) show an improvement of 2.3% and 4.6%, which demonstrates the validity of the proposed GCWNet for multiscale objects. The results of the ablation experiment illustrated our motivation brilliantly.

In addition, to demonstrate optimal GCAM settings, we drop a feature layer from the GCAM, and the results of the ablation study are shown in Table III. It is observed that only three layers of feature weaving have lower results than GCAM, and lower layer features have more influence on the results. We believe it is because lower level features contain richer global context information and contribute more to GCAM. Although the baseline is more effective in the three categories of tennis court (TC), SH, and helicopter (HC), GCWNet outperforms the baseline in all the three categories, which also shows that FRM and loss have compensatory properties for GCAM.

D. Comparison Study

To comprehensively evaluate the superiority of the proposed method, we evaluate GCWNet against the currently reported object detection methods, which is shown in Table IV. The comparison algorithms are divided into two groups, the first one is the one-stage algorithms, including SSD [30], RetinaNet [31], and YOLOv4 [29], while the other one is the two-stage algorithms, including Faster RCNN [36], Faster RCNN with FPN [37], PANet [39], and CDD-Net [63]. From Table IV, GCWNet is apparently superior to other methods, for instance, by 7.1% and 1.4% of mAP, respectively. over the one-stage framework YOLOv4 and the two-stage framework CDD-Net. Moreover, the model parameters and the efficiency of the proposed method are shown in Table V, which indicates that

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:58:37 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

PERFORMANCE EVALUATION OF SIX DIFFERENT METHODS ON THE DOTA-v1.5 DATASET

Methods	**mAP**	BD	TC	SH	BC	PL	GTF	HA	BR	SV	SBF	LV	HC	SP	ST	RA	CC
RetinaNet [31]	33.5	44.5	75.1	33.4	30.8	76.0	32.5	35.8	32.6	10.7	13.0	33.3	0.2	43.9	31.2	42.4	0
SSD [30]	36.3	38.6	81.7	30.6	42.0	78.1	27.4	46.9	14.5	13.5	20.7	39.5	30.4	39.3	43.7	33.6	0
YOLOv4 [29]	55.6	61.7	88.3	**79.5**	55.6	85.2	35.2	69.8	32.6	**37.0**	34.4	**64.0**	67.6	58.5	**64.8**	54.2	0.7
Faster RCNN [36]	51.8	63.4	81.3	32.5	68.4	70.0	63.3	59.1	55.5	22.7	57.6	44.5	51.5	55.1	36.9	49.8	16.9
Faster RCNN with FPN [37]	57.3	70.1	86.0	40.3	69.4	78.6	68.5	59.5	55.1	23.7	61.1	45.4	68.3	64.5	46.4	56.2	24.4
PANet [39]	61.2	74.1	89.6	58.4	67.0	**85.9**	64.5	67.9	51.5	27.7	63.4	56.2	71.3	**73.4**	61.3	59.2	7.6
CDD-Net [63]	61.3	74.7	**89.8**	49.2	71.4	81.4	70.1	69.9	55.3	25.3	65.6	51.5	71.3	60.4	53.3	58.2	32.7
GCWNet	**62.7**	**77.3**	89.7	53.7	**71.4**	81.3	**70.8**	**70.3**	**55.7**	26.0	**65.7**	54.2	**72.2**	69.0	52.6	**59.6**	**33.6**

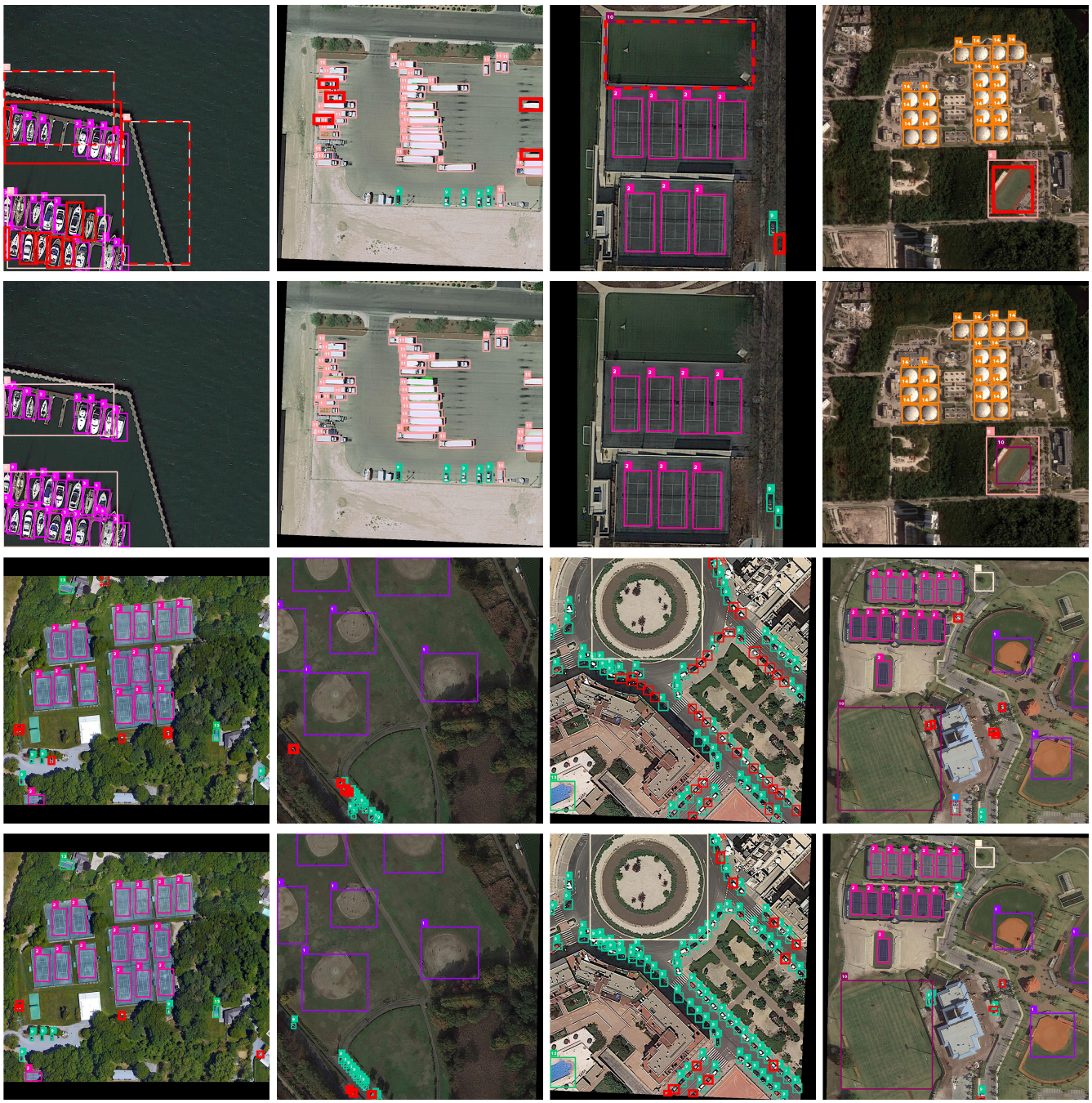


Fig. 4. Some results of remote sensing image detection on the DOTA-v1.5 dataset. The first and third lines display the detection baseline. The second and fourth lines show the detection of the proposed GCWNet, where all the objects in the second row are accurately detected and the instances in the fourth row have some false negative samples. Complementarily, true positive detection results are indicated by different colored boxes and the category number is provided in the top-left corner. The red boxes and red dashed boxes refer to false negative and false positive instances, respectively.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:58:37 UTC from IEEE Xplore. Restrictions apply.

TABLE V
PARAMETERS AND EFFICIENCY OF DIFFERENT MODELS

Methods	Backbone	Parameters/M	Test time/ms
RetinaNet [31]	ResNet101	52.8	59
SSD [30]	VGG16	22.6	34
YOLOv4 [29]	DarkNet53	32.6	16
Faster RCNN [36]	ResNet101	45.1	45
Faster RCNN with FPN [37]	ResNet101	62.2	49
PANet [39]	ResNet101	75.7	107
CDD-Net [63]	ResNet101	63.4	50
GCWNet	ResNet101	69.3	53

the inference speed of GCWNet is only 53 ms. Therefore, the superiority of GCWNet is obviously demonstrated.

1) Comparative One-Stage Algorithms: There is nearly incredible overwhelming superiority of our model compared with the one-stage algorithms. In particular, the proposed method achieves 29.2% and 26.4% enhancement compared with RetinaNet and SSD, respectively, and outperforms both the methods in terms of accuracy in each category overall. mAP is increased by 7.1% in comparison with the recently proposed YOLOv4. YOLOv4 achieves optimal detection for three categories due to the fact that the method is sensitive to small-scale objects. However, for large-scale instances such as ground track field (GTF), the proposed method is 35.6% higher than YOLOv4. In addition, for container crane (CC), the one-stage methods perform poorly, not only for the inherent shortcomings of the one-stage approach but also owing to the relatively few numbers of instances in the category. Consequently, compared with the one-stage method, GCWNet is more preferable and capable of accommodating both large-scale and small-scale objects, while providing some refinement to the category imbalanced. In addition, it is evident from Table V that the model parameters of the one-stage algorithms are less and the inference speed is faster compared with GCWNet. However, the 53-ms inference speed of our model is completely acceptable while the parameters are increased.

2) Comparative Two-Stage Algorithms: As shown in Table IV, the two-stage methods significantly perform better than the one-stage methods, and our model achieves the optimal effect in the two-stage methods. In fact, the Faster RCNN raises the map by more than 15% over RetinaNet and SSD, and the baseline mAP is greater than the state-of-the-art one-stage method, YOLOv4. In contrast to Faster RCNN and Faster RCNN with FPN, our model achieves a 10.9% and 5.4% boost, respectively, while all the categories show various degrees of accuracy progress. GCWNet improves mAP over PANet by 1.5% and outperforms PANet in ten categories, indicating that GCWNet is more appropriate for remote sensing object detection. Compared with CDD-Net designed for DOTA-v1.5, the proposed model attains a 1.4% improvement and surpasses the performance in 13 categories. Moreover, for CC with few instances, the mAP of other methods could reach at most 24.4% and 32.7%; however, the proposed method increases by 9.2% and 0.9%–33.6%. The fact reaffirms the improvement of GCWNet for imbalanced issue. It is evident that GCWNet is capable of performing the optimal detection performance so far compared with the two-stage methods, which illustrates the desirability and feasibility of global context aggregation. Notably, our model has an 11.4% increase in parameters compared with baseline, yet the test time is only 4 ms slower.

3) Visual Results: Fig. 4 shows the visualization of the detection results of the proposed GCWNet and the baseline model Faster RCNN with FPN on the DOTA-v1.5 dataset. To provide as many representative results as possible and to facilitate comparisons, the first and third lines present the results of the baseline model, while the second and fourth lines present the results of the GCWNet model. It is feasible to observe from Fig. 4 that the Faster RCNN with FPN tends to produce false positives in different scenes, such as recognizing similar looking patches as HAs, SBF, SVs, and plane (PL) in the first, third, fifth, and eighth examples, respectively. Meanwhile, owing to the dense distribution of instances in remote sensing images, Faster RCNN with FPN tends to omit some objects in dense scenes, such as SHs, large vehicles (LVs), and SVs in the first, second, sixth, and seventh sample images. It is also observed that the baseline model misses detection for multiscale objects, especially small ones, such as SVs in the third, fifth, and eighth sample images and SBF in the fourth sample image. In contrast, the proposed GCWNet is able to properly detect the densely distributed multiscale objects in various scenes with minimal false positives. It suggests that global context features and feature refinement effectively guide feature generation and facilitate the detection of multiscale objects.

On the other hand, the proposed GCWNet has detection defects in specific cases, as shown in the fourth row of Fig. 4. For densely distributed small objects, the proposed model suffers from few instances of missed detection, as for the sample images in the second and third columns, which is mainly attributed to the proximity between the objects and the difficulty in distinguishing between the vehicles and the background. Moreover, GCWNet also misses for extremely small objects, such as the sample images in the first and fourth columns, which is an inherent drawback of the baseline model, meaning that the anchor box is not possible to achieve a particularly small scale. We will continuously research to address these issues in our future work.

V. CONCLUSION

In this article, a novel global context-weaving object detection network called GCWNet is meticulously designed, which accurately locates and identifies instances in remote sensing images. GCAM enables the extraction of global context features by aggregating information from different layers, hence boosting the detection of dense objects. FRM is implemented to refine the feature representation and enhance the suitability of the network for multiscale objects with convolution of different receptive fields. In addition, we introduce focal loss and balanced L1 loss instead of cross-entropy loss and smoothed L1 loss in the baseline, respectively. The imbalance issue in the training process is alleviated by replacing the loss functions in the classification and regression branches. The extensive and sufficient experimental results on the DOTA-v1.5 dataset verify that GCWNet has remarkably outstanding object detection performance in comparison to the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:58:37 UTC from IEEE Xplore. Restrictions apply.

state-of-the-art algorithms. We will investigate more productive strategies to tackle small objects and improve the perception of small objects. Furthermore, we will research more contextual applications in object detection to further boost computational efficiency and enhance detection performance.

REFERENCES

[1] G. Cheng and J. Han, "A survey on object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 117, pp. 11–28, Jul. 2016.

[2] X. Li, Y. Yuan, and Q. Wang, "Hyperspectral and multispectral image fusion via nonlocal low-rank tensor approximation and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 550–562, Jan. 2021.

[3] L. Zhang, M. Lan, J. Zhang, and D. Tao, "Stagewise unsupervised domain adaptation with adversarial self-training for road segmentation of remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[4] K. Liu, J. Huang, M. Xu, M. Perc, and X. Li, "Density saliency for clustered building detection and population capacity estimation," *Neurocomputing*, vol. 458, pp. 127–140, Oct. 2021.

[5] Q. An, Z. Pan, L. Liu, and H. You, "DRBox-v2: An improved detector with rotatable boxes for target detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8333–8349, Nov. 2019.

[6] X. Li, M. Chen, F. Nie, and Q. Wang, "Locality adaptive discriminant analysis," in *Proc. 26th Int. Joint Conf. Artif. Intell.*, Aug. 2017, pp. 2201–2207.

[7] Z. Dong, M. Wang, Y. Wang, Y. Zhu, and Z. Zhang, "Object detection in high resolution remote sensing imagery based on convolutional neural networks with suitable object scale features," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 2104–2114, Mar. 2020.

[8] Q. Li, L. Mou, Q. Xu, Y. Zhang, and X. X. Zhu, "R3-Net: A deep network for multioriented vehicle detection in aerial images and videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 5028–5042, Jul. 2019.

[9] Y. Yu, X. Yang, J. Li, and X. Gao, "A cascade rotated anchor-aided detector for ship detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5600514.

[10] Y. Qi, S. Zhang, F. Jiang, H. Zhou, D. Tao, and X. Li, "Siamese local and global networks for robust face tracking," *IEEE Trans. Image Process.*, vol. 29, pp. 9152–9164, 2020.

[11] G. Sun *et al.*, "Deep fusion of localized spectral features and multi-scale spatial features for effective classification of hyperspectral images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 91, Sep. 2020, Art. no. 102157.

[12] X. Li, D. Song, and Y. Dong, "Hierarchical feature fusion network for salient object detection," *IEEE Trans. Image Process.*, vol. 29, pp. 9165–9175, 2020.

[13] X. Liang, J. Zhang, L. Zhuo, Y. Li, and Q. Tian, "Small object detection in unmanned aerial vehicle images using feature fusion and scaling-based single shot detector with spatial context analysis," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 6, pp. 1758–1770, Jun. 2020.

[14] Q. Yao, X. Hu, and H. Lei, "Multiscale convolutional neural networks for geospatial object detection in VHR satellite images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 1, pp. 23–27, Jan. 2021.

[15] G. Sun *et al.*, "SpaSSA: Superpixelwise adaptive SSA for unsupervised spatial–spectral feature extraction in hyperspectral image," *IEEE Trans. Cybern.*, early access, Sep. 9, 2021, doi: 10.1109/TCYB.2021.3104100.

[16] C. Li *et al.*, "Learning object-wise semantic representation for detection in remote sensing imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2019, pp. 20–27.

[17] L. Du, L. Li, D. Wei, and J. Mao, "Saliency-guided single shot multibox detector for target detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3366–3376, May 2020.

[18] X. Lu, Y. Zhong, Z. Zheng, and L. Zhang, "GAMSNet: Globally aware road detection network with multi-scale residual learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 340–352, May 2021.

[19] X. Zhang, G. Wang, P. Zhu, T. Zhang, C. Li, and L. Jiao, "GRS-det: An anchor-free rotation ship detector based on Gaussian-mask in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3518–3531, Apr. 2021.

[20] X. Yang *et al.*, "SCRDet: Towards more robust detection for small, cluttered and rotated objects," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8232–8241.

[21] G. Shi, J. Zhang, J. Liu, C. Zhang, C. Zhou, and S. Yang, "Global context-augmented objection detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10604–10617, Dec. 2021.

[22] J. Wang, G. Zhang, K. Zhang, Y. Zhao, Q. Wang, and X. Li, "Detection of small aerial object using random projection feature with region clustering," *IEEE Trans. Cybern.*, early access, Sep. 29, 2020, doi: 10.1109/TCYB.2020.3018120.

[23] W. Guo, W. Li, Z. Li, W. Gong, J. Cui, and X. Wang, "A slimmer network with polymorphic and group attention modules for more efficient object detection in aerial images," *Remote Sens.*, vol. 12, no. 22, p. 3750, Nov. 2020.

[24] C. Xu, C. Li, Z. Cui, T. Zhang, and J. Yang, "Hierarchical semantic propagation for object detection in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4353–4364, Jun. 2020.

[25] L. Shi, L. Kuang, X. Xu, B. Pan, and Z. Shi, "CANet: Centerness-aware network for object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[26] P. Wang, X. Sun, W. Diao, and K. Fu, "FMSSD: Feature-merged single-shot detection for multiscale objects in large-scale remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3377–3390, Dec. 2020.

[27] J. Cao, Y. Pang, S. Zhao, and X. Li, "High-level semantic networks for multi-scale object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 10, pp. 3372–3386, Oct. 2020.

[28] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[29] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "YOLOv4: Optimal speed and accuracy of object detection," 2020, *arXiv:2004.10934*.

[30] W. Liu *et al.*, "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis.*, Cham, Switzerland: Springer, 2016, pp. 21–37.

[31] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollar, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[32] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 580–587.

[33] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 25. Stateline, NV, USA, Dec. 2012, pp. 1097–1105.

[34] J. R. R. Uijlings, K. E. A. van de Sande, T. Gevers, and A. W. M. Smeulders, "Selective search for object recognition," *Int. J. Comput. Vis.*, vol. 104, no. 2, pp. 154–171, Apr. 2013.

[35] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[36] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[37] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.

[38] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 2961–2969.

[39] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8759–8768.

[40] J. Wang, Z. Ma, F. Nie, and X. Li, "Progressive self-supervised clustering with novel category discovery," *IEEE Trans. Cybern.*, early access, Apr. 20, 2021, doi: 10.1109/TCYB.2021.3069836.

[41] H. Zhou, L. Wei, C. P. Lim, D. Creighton, and S. Nahavandi, "Robust vehicle detection in aerial images using bag-of-words and orientation aware scanning," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 12, pp. 7074–7085, Dec. 2018.

[42] J. Tu, F. Gao, J. Sun, A. Hussain, and H. Zhou, "Airport detection in SAR images via salient line segment detector and edge-oriented region growing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 314–326, 2021.

[43] Z. Fang, J. Ren, H. Sun, S. Marshall, J. Han, and H. Zhao, "SAFDet: A semi-anchor-free detector for effective detection of oriented objects in aerial images," *Remote Sens.*, vol. 12, no. 19, p. 3225, Oct. 2020.

[44] D. Xu and Y. Wu, "MRFF-YOLO: A multi-receptive fields fusion network for remote sensing target detection," *Remote Sens.*, vol. 12, no. 19, p. 3118, Sep. 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:58:37 UTC from IEEE Xplore. Restrictions apply.

[45] B. Hou, Z. Ren, W. Zhao, Q. Wu, and L. Jiao, "Object detection in high-resolution panchromatic images using deep models and spatial template matching," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 956–970, Feb. 2020.
[46] Q. Guo, H. Wang, and F. Xu, "Scattering enhanced attention pyramid network for aircraft detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7570–7587, Sep. 2021.
[47] Y. Zhao, L. Zhao, B. Xiong, and G. Kuang, "Attention receptive pyramid network for ship detection in SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 2738–2756, 2020.
[48] Z. Cui, Q. Li, Z. Cao, and N. Liu, "Dense attention pyramid networks for multi-scale ship detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8983–8997, Nov. 2019.
[49] J. Fu, X. Sun, Z. Wang, and K. Fu, "An anchor-free method based on feature balancing and refinement network for multiscale ship detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1331–1344, Feb. 2021.
[50] G. Huang, Z. Wan, X. Liu, J. Hui, Z. Wang, and Z. Zhang, "Ship detection based on squeeze excitation skip-connection path networks for optical remote sensing images," *Neurocomputing*, vol. 332, pp. 215–223, Mar. 2019.
[51] K. Fu *et al.*, "A ship rotation detection model in remote sensing images based on feature fusion pyramid network and deep reinforcement learning," *Remote Sens.*, vol. 10, no. 12, p. 1922, Nov. 2018.
[52] L. Zhang, Y. Wang, and Y. Huo, "Object detection in high-resolution remote sensing images based on a hard-example-mining network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8768–8780, Oct. 2021.
[53] Z. Yuan, Z. Liu, C. Zhu, J. Qi, and D. Zhao, "Object detection in remote sensing images via multi-feature pyramid network with receptive field block," *Remote Sens.*, vol. 13, no. 5, p. 862, Feb. 2021.
[54] X. Wu, D. Sahoo, and S. C. H. Hoi, "Recent advances in deep learning for object detection," *Neurocomputing*, vol. 396, pp. 39–64, Jul. 2020.
[55] X. Li, M. Chen, F. Nie, and Q. Wang, "A multiview-based parameter free framework for group detection," in *Proc. AAAI Conf. Artif. Intell.*, 2017, pp. 4147–4153.
[56] Z.-M. Chen, X. Jin, B. Zhao, X.-S. Wei, and Y. Guo, "Hierarchical context embedding for region-based object detection," in *Proc. Eur. Conf. Comput. Vis.*, Cham, Switzerland: Springer, 2020, pp. 633–648.
[57] J. Nie, R. M. Anwer, H. Cholakkal, F. S. Khan, Y. Pang, and L. Shao, "Enriched feature guided refinement network for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9537–9546.
[58] J. Leng, Y. Ren, W. Jiang, X. Sun, and Y. Wang, "Realize your surroundings: Exploiting context information for small object detection," *Neurocomputing*, vol. 433, pp. 287–299, Apr. 2021.
[59] Z. Chen, J. Zhang, and D. Tao, "Recursive context routing for object detection," *Int. J. Comput. Vis.*, vol. 129, no. 1, pp. 142–160, 2020.
[60] C. Tao, L. Mi, Y. Li, J. Qi, Y. Xiao, and J. Zhang, "Scene context-driven vehicle detection in high-resolution aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7339–7351, Oct. 2019.
[61] C. Chen, W. Gong, Y. Chen, and W. Li, "Object detection in remote sensing images based on a scene-contextual feature pyramid network," *Remote Sens.*, vol. 11, no. 3, p. 339, Feb. 2019.
[62] Y. Gong *et al.*, "Context-aware convolutional neural network for object detection in VHR remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 34–44, Jan. 2019.
[63] Y. Wu, K. Zhang, J. Wang, Y. Wang, Q. Wang, and Q. Li, "CDD-Net: A context-driven detection network for multiclass object detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[64] G. Zhang, S. Lu, and W. Zhang, "CAD-Net: A context-aware detection network for objects in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 10015–10024, Aug. 2019.
[65] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[66] K. Zhang, Y. Wu, J. Wang, Y. Wang, and Q. Wang, "Semantic context-aware network for multiscale object detection in remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[67] K. Sun, B. Xiao, D. Liu, and J. Wang, "Deep high-resolution representation learning for human pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5693–5703.
[68] X. Ding, Y. Guo, G. Ding, and J. Han, "ACNet: Strengthening the kernel skeletons for powerful CNN via asymmetric convolution blocks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1911–1920.
[69] G.-S. Xia *et al.*, "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.



Yulin Wu (Student Member, IEEE) received the B.S. and M.S. degrees from Northwestern Polytechnical University, Xi'an, China, in 2014 and 2017, respectively, where she is currently pursuing the Ph.D. degree in guidance, navigation, and control.

Her research interests include machine learning, signal processing, and object detection.



Ke Zhang received the Ph.D. degree from Northwestern Polytechnical University, Xi'an, China, in 1998.

He is currently the Associate Dean of the School of Astronautics and a Full Professor with the National Key Laboratory of Aerospace Flight Dynamics, Northwestern Polytechnical University. His research interests include image and signal processing and intelligent perception.



Jingyu Wang (Member, IEEE) received the Ph.D. degree in signal, image, and automatic from Université Paris-Est, Paris, France, in 2015.

He is currently a Full Professor with the School of Astronautics and the School of Artificial Intelligence, OPtics and ElectroNics (iOPEN), Northwestern Polytechnical University, Xi'an, China. His research interests include information processing, computer vision, and intelligent perception.



Yezi Wang received the B.S. and M.S. degrees in control science and engineering from Northwestern Polytechnical University, Xi'an, China, in 2018 and 2021, respectively.

She is currently an Assistant Engineer with the Structural Health Monitoring and Intelligent Structures Laboratory, Aircraft Strength Research Institute of China, Xi'an. Her research interests include computer vision, machine learning, and structural health monitoring.



Qi Wang (Senior Member, IEEE) received the B.E. degree in automation and the Ph.D. degree in pattern recognition and intelligent systems from the University of Science and Technology of China, Hefei, China, in 2005 and 2010, respectively.

He is currently a Professor with the School of Artificial Intelligence, OPtics and ElectroNics (iOPEN), Northwestern Polytechnical University, Xi'an, China. His research interests include computer vision and pattern recognition.

Xuelong Li (Fellow, IEEE) is currently a Full Professor with the School of Artificial Intelligence, OPtics and ElectroNics (iOPEN), Northwestern Polytechnical University, Xi'an, China.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:58:37 UTC from IEEE Xplore. Restrictions apply.